

TCAN1051HVD



DevLab: TCAN1051HVD CAN
Transceiver Module

Professional electronic component

v1.0
2025-10-02
Rev. A

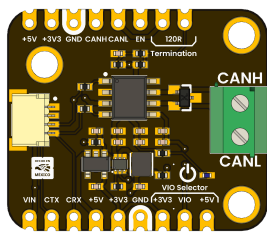
PRODUCT OVERVIEW

The TCAN1051HVD is a high-speed CAN transceiver designed for automotive and industrial applications. It provides a robust interface between a CAN controller and the physical CAN bus, ensuring reliable data transmission in harsh environments. This module is ideal for use in electric vehicles, industrial automation, and other applications requiring reliable CAN communication.

Feature / Component	Detailed Description
Communication	Integrated high-speed CAN (Controller Area Network) interface, ensuring robust and reliable data transfer in demanding environments.
Operating Voltage	Supports both 3.3V and 5V logic levels, making it adaptable to a wide range of microcontrollers and external systems.
Input Port	JST 1.0 mm 4-pin connector providing a compact and secure interface for supplying power and CAN signals.
IC1 – TCAN1051HVD	Automotive-grade CAN transceiver IC, designed for high noise immunity, ESD protection, and reliable long-distance communication.
IC2 – AP2112K	3.3V low dropout (LDO) regulator, delivering a clean and stable voltage supply to sensitive components.
IC3 – TPS61023	High-efficiency 5V boost converter, enabling stable power delivery even from lower input voltages.
L1 – Power LED	On-board LED indicator that provides instant visual feedback of power status.
JP1 / JP2	Dual rows of 2.54 mm castellated holes (side 1 and 2), allowing flexible mounting, soldering, or integration into custom boards.
J1 – CAN Data Port	Dedicated JST 1.0 mm pitch connector optimized for reliable CAN bus data line interfacing.
J2 – Screw Terminal	Sturdy terminal block that allows secure, tool-based connection to the CAN bus wiring.
SB1 – Logic Voltage Selector	Configurable solder bridge for selecting the appropriate logic voltage (VIO), ensuring compatibility with the host system.

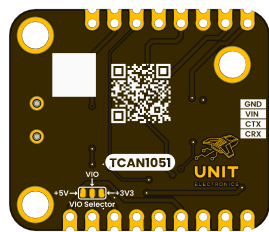
PRODUCT VIEWS

TOP VIEW



Component placement and connectors

BOTTOM VIEW



Underside components and connections

KEY TECHNICAL SPECIFICATIONS



CONNECTIVITY

Primary Interface: **GPIO (Interrupt)**
Connector Type: **JST 4-pin 1.0mm**
Logic Levels: **VCC-referenced (2V – 5.5V tolerant)**

PIN CONFIGURATION

FUNCTION		NOTES
Power	Supply	3.3V or 5V

COMPONENT REFERENCE

REF.	DESCRIPTION
IC1	CAN Transceiver
IC2	AP2112K 3.3V Regulator
IC3	TPS61023 5V Regulator
L1	Power On LED
JP1	2.54 mm Castellated Holes
JP2	2.54 mm Castellated Holes
J1	JST 1 mm pitch for CAN Data
J2	Terminal Screw Block for CAN Bus

KEY FEATURES

Touch-only sensing

No physical press required – reacts to proximity of a finger.

Auto-calibration

Compensates for environmental changes and drift.

On-board pull-up/down

Ensures clean digital output.

JST PH-2.0 connector

Quick-disconnect cable interface.

Fast response

< 80 ms touch detection time.

Selectable modes

Momentary or toggle output (via solder-jumper on the board).

Mounting holes

Two M3 screw holes for easy panel integration.

Key Applications

Automotive systems, Robotics, Home automation, IoT devices

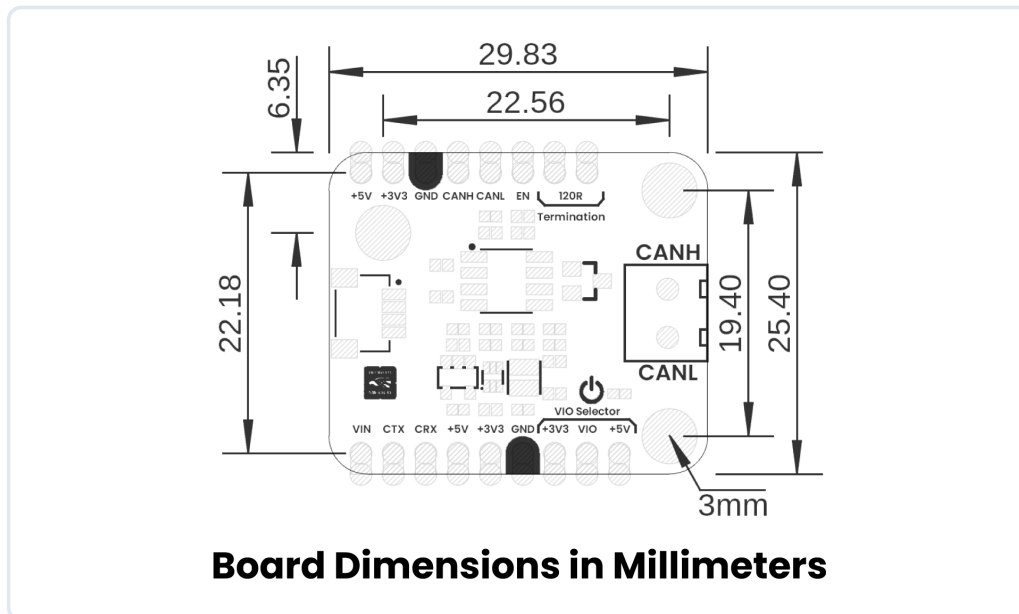
ADDITIONAL TECHNICAL INFORMATION

OVERVIEW

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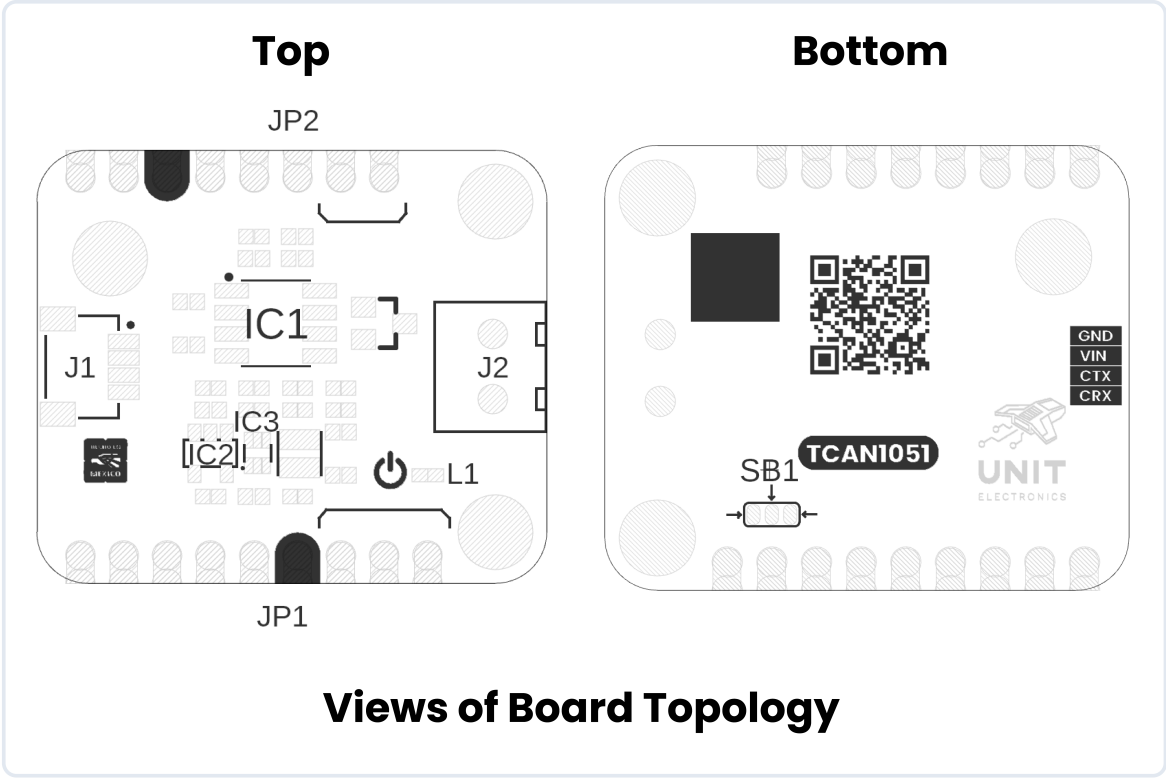
HARDWARE DOCUMENTATION

MECHANICAL DIMENSIONS



Physical dimensions and mounting specifications (measurements in millimeters)

SYSTEM TOPOLOGY



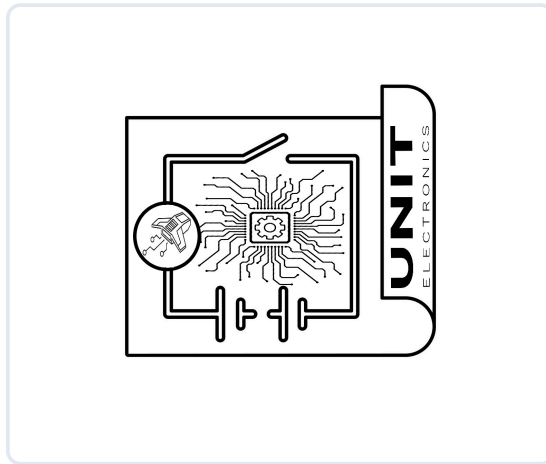
Connection topology and system integration diagram

[Click image to open in full size](#)

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SB1	Solder Bridge for VIO

CIRCUIT SCHEMATIC



Complete circuit schematic showing all component connections

[View Complete Schematic PDF](#)

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION		NOTES
Power Supply	3.3V or 5V	
Ground	Common ground for all components	

PIN CONFIGURATION LAYOUT

Physical connector layout and pin positioning



Pin Configuration Layout

Complete pin configuration diagram showing all connectors, pin assignments, and electrical connections for proper integration

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Professional Technical Datasheet

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